

Mathematical Foundations of Reinforcement Learning and Online Decision Making

Lecture 3

Markov Decision Processes (MDPs) and Dynamic Programming

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Overview

- 1 Markov Reward Process
- 2 MRP - Discounted Problems
- 3 MRP - Shortest Path Problems
- 4 MDP - Discounted Problems
- 5 MDP - Shortest Path Problems

Markov Reward Process

A Markov chain is a pair $(\mathcal{S}, P, R)$ of

- 1 **States:** $\mathcal{S}$
- 2 **Transitions:** $P : \mathcal{S} \times \mathcal{S} \rightarrow [0, 1]$ where

$$\sum_{s' \in \mathcal{S}} P(s, s') = 1 \quad \text{for all } s \in \mathcal{S}.$$

- 3 **Reward function:** $R : \mathcal{S} \rightarrow \mathbb{R}$

Markov Reward Process - Trajectory

A trajectory (realization) of $(\mathcal{S}, P, R)$ is a sequence

$$S_0, R_0, S_1, R_1, S_2, R_2 \dots, S_t, R_t \dots$$

where

$$\begin{aligned} P(S_{t+1}=s' | S_t=s, S_{t-1}=s_{t-1}, \dots, S_0=s_0) &= P(S_{t+1}=s' | S_t=s) \\ &= P(s, s') \end{aligned}$$

and

$$R_t = R(S_t) \quad [\text{or } R_t \sim R(S_t)]$$

Markov Reward Process - Value of a State

Criterion 1: Discounted Problem

$$V(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(S_t) \mid S_0 = s \right]$$

Criterion 2: Stochastic Shortest Path Problem

$$V(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} R(S_t) \mid S_0 = s \right]$$

Criterion 3: Average Cost per Stage Problem

$$V(s) = \mathbb{E} \left[\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^T R(S_t) \mid S_0 = s \right]$$

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Markov Reward Process - Discounted Problem

How to compute the value function?

Markov Reward Process - Discounted Problem

How to compute the value function?

Bellman Equation

$$\begin{aligned} V(s) &= \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(S_t) \mid S_0 = s \right] \\ &= R(s) + \gamma \sum_{s' \in \mathcal{S}} \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t R(S_t) \mid S_1 = s' \right] \\ &= R(s) + \gamma \sum_{s' \in \mathcal{S}} P(s, s') V(s') \end{aligned}$$

Markov Reward Process - Discounted Problem

Bellman Equation in matrix form

$$v = R + \gamma P v$$

Can compute v by solving the linear system

$$(I - \gamma P)v = R.$$

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Is there a unique solution?

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Can compute v by solving the linear system

$$(I - \gamma P)v = R.$$

Is there a unique solution?

Yes, since $|\lambda(\gamma P)| \leq \gamma < 1$, so we have

$$v = (I - \gamma P)^{-1} R$$

Markov Reward Process - Discounted Problem

Bellman Equation in matrix form

$$v = R + \gamma P v$$

Can also compute v by Dynamic Programming:

$$v^{k+1} = T(v^k) := R + \gamma P v^k$$

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Does this algorithm converge?

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Can also compute v by Dynamic Programming:

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Does this algorithm converge?

$$\|T(v) - T(w)\|_{\infty} \leq \|\gamma P(v - w)\|_{\infty} \leq \gamma \|P\|_{\infty} \|v - w\|_{\infty} = \gamma \|v - w\|_{\infty}$$

Markov Reward Process - Discounted Problem

Bellman Equation in matrix form

$$v = R + \gamma P v$$

Homework Question: When is solving Dynamic Programming more efficient than solving a Linear System?

Note: DP solution is the base for many RL-algorithms.

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Markov Reward Process - Shortest Path Problem

$$V(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} R(X_t) \mid S_0 = s \right]$$

Does this value always exist?

Markov Reward Process - Shortest Path Problem

$$V(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} R(X_t) \mid S_0 = s \right]$$

Does this value always exist?

Generally no. Consider a system with one state and positive reward, then $V(s) = \infty$.

Markov Reward Process - Shortest Path Problem

$$V(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} R(S_t) \mid S_0 = s \right]$$

When does this value always exist?

Markov Reward Process - Shortest Path Problem

$$V(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} R(S_t) \mid S_0 = s \right]$$

When does this value always exist?

Stochastic Shortest Path

Consider $S = \{0, 1, \dots, n\}$, where 0 is a cost free terminal state:

$$R(0) = 0, \quad \text{and} \quad P(0, 0) = 1.$$

Moreover, for any $s \in S$ there is i such that $P^i(s, 0) > 0$.

Markov Reward Process - Shortest Path Problem

How to compute this value function?

Markov Reward Process - Shortest Path Problem

How to compute this value function?

Bellman Equation

$$\begin{aligned} V(s) &= \mathbb{E} \left[\sum_{t=0}^{\infty} R(S_t) \mid S_0 = s \right] \\ &= R(s) + \sum_{s' \in \mathcal{S}} \mathbb{E} \left[\sum_{t=1}^{\infty} R(S_t) \mid S_1 = s' \right] \\ &= R(s) + \sum_{s' \in \mathcal{S}} P(s, s') V(s') \end{aligned}$$

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Can compute v by solving the linear system

$$(I - P)v = R.$$

Is there a unique solution?

No, P has 1 as a singular eigenvalue (Perron–Frobenius theorem).

Markov Reward Process - Shortest Path Problem

Bellman Equation in matrix form

$$v = R + Pv$$

Under the assumptions of SP problems, there is a unique solution under the constraint $v(0) = 0$.

Homework: Prove this and compute the solution.

Markov Reward Process - Shortest Path Problem

Bellman Equation in matrix form

$$v = R + Pv$$

The DP algorithm can also be used

$$v^{k+1} = R + Pv^k$$

provided that we fix $v^k(0) = 0$ at every iteration k .

Homework: Prove that this DP algorithm is contractive, and hence converges to its fixed point.

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Markov Decision Process

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- 1 **States:** $\mathcal{S}$
- 2 **Actions:** $\mathcal{A}$
- 3 **Transitions:** $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ where

$$\sum_{s' \in \mathcal{S}} P(s'|s, a) = 1 \quad \text{for all } (s, a) \in \mathcal{S} \times \mathcal{A}.$$

- 4 **Reward function:** $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$

MDP - Trajectory

A trajectory (realization) of (S, P, R) is a sequence

$$S_0, A_0, R_0, S_1, A_1, R_1, S_2, A_2, R_2 \dots, S_t, A_t, R_t \dots$$

where

$$\begin{aligned} P(S_{t+1}=s' | S_t=s, A_t=a_t, \dots, A_0=a_0) &= P(S_{t+1}=s' | S_t=s, A_t=a) \\ &= P(s, s') \end{aligned}$$

and

$$R_t = R(S_t, A_t) \quad \left[\text{or } R_t \sim R(S_t, A_t) \right]$$

MDP - Policy

Want to find a good policy:

$$\pi : \mathcal{S} \rightarrow \mathcal{A} \quad (\text{Deterministic})$$

$$p_{\pi}(a|s) : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1] \quad (\text{Stochastic})$$

For a fixed policy π we have a MRP with transition matrix P_{π} where

$$P_{\pi}(s, s') = P(s'|s, \pi(s))$$

MDP - Value of a State

Discounted Problem

$$V_{\pi}(s) = \mathbb{E}_{S_t \sim P_{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t R(S_t) \mid S_0 = s \right]$$

Can we find the optimal policy?

$$\max_{\pi} V_{\pi}(s)$$

MDP - Value of a State

Can we find the optimal policy?

$$\max_{\pi} V_{\pi}(s)$$

Properties of optimal policy

For a MDPs we have:

- the optimal policy is Markovian
- Same π optimal for all $s \in \mathcal{S}$
- There exist a deterministic policy that is optimal.

MDP - Policy Improvement

Can we find the optimal policy?

$$\max_{\pi} V_{\pi}(s)$$

Algorithms

- Policy iteration
- Value iteration
- Linear programming

MDP - Q function

For given policy π the Q-function

$$\begin{aligned} Q_{\pi}(s, a) &= \mathbb{E}_{\pi} \left(\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t) \mid S_0 = s, A_0 = a \right) \\ &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V_{\pi}(s') \\ &= r(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s' | s, a) V_{\pi}(s') \end{aligned}$$

MDP - Policy Iteration

- **Policy Evaluation:** Compute V_{π_k} (or Q_{π_k}) using the methods of MRP.
- **Policy improvement:**

$$\pi_{k+1}(s) = \operatorname{argmax}_a Q(s, a)$$

- $\pi_{k+1} \geq \pi_k$. Why?
- $\pi_{k+1} = \pi_k$ only if π_k is optimal policy. Why?
- Will converge in finite number of iterations. Why?

MDP - Value Iteration

Compute directly the optimal Q-table

$$Q(s, a) = \max_{\pi} Q_{\pi}(s, a)$$

Can be computed by with the optimal Bellmann operator:

$$\begin{aligned}\mathcal{T}(Q)(s, a) &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left(\max_{a' \in \mathcal{A}} Q(s', a') \right) \\ &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left(\max_{a' \in \mathcal{A}} Q(s', a') \right)\end{aligned}$$

Can compute Q as

$$Q^{t+1}(s, a) = \mathcal{T}(Q^t)(s, a)$$

Homework: Prove that $\mathcal{T}$ is a contraction with fixed point Q .

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MDP - Shortest Path Problem

Goal of the Shortest Path Problem

$$V_{\pi}(s) = \mathbb{E}_{S_t \sim P_{\pi}} \left[\sum_{t=0}^{\infty} R(S_t) \mid S_0 = s \right]$$

Can we find the optimal policy?

$$\max_{\pi} V_{\pi}(s)$$

MDP - Shortest Path Problem

Can we find the optimal policy?

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MDP - Shortest Path Problem

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